LABORATORY REPORT

Contract Number: PSL26/3595

Report Date: 27 April 2026

Client's Reference:

Client Name: Sirius Leeds
4245 Park Approach
Century Way
Thorpe Park
Leeds
LS15 8GB

For the attention of: Ami Cooper

Contract Title: Land off Tinker Lane, Rufforth, York, YO23 3QA

Date Received: 24/4/2026
Date Commenced: 24/4/2026
Date Completed: 27/4/2026

Notes: **Opinions and Interpretations are outside the UKAS Accreditation**

A copy of the Laboratory Schedule of accredited tests as issued by UKAS is attached to this report. This certificate is issued in accordance with the accreditation requirements of the United Kingdom Accreditation Service. The results reported herein relate only to the material supplied to the laboratory. This certificate shall not be reproduced other than in full, without the prior written approval of the laboratory.

Checked and Approved Signatories:

A Watkins (Managing Director)	R Berriman (Associate Director)	S Royle (Laboratory Manager)
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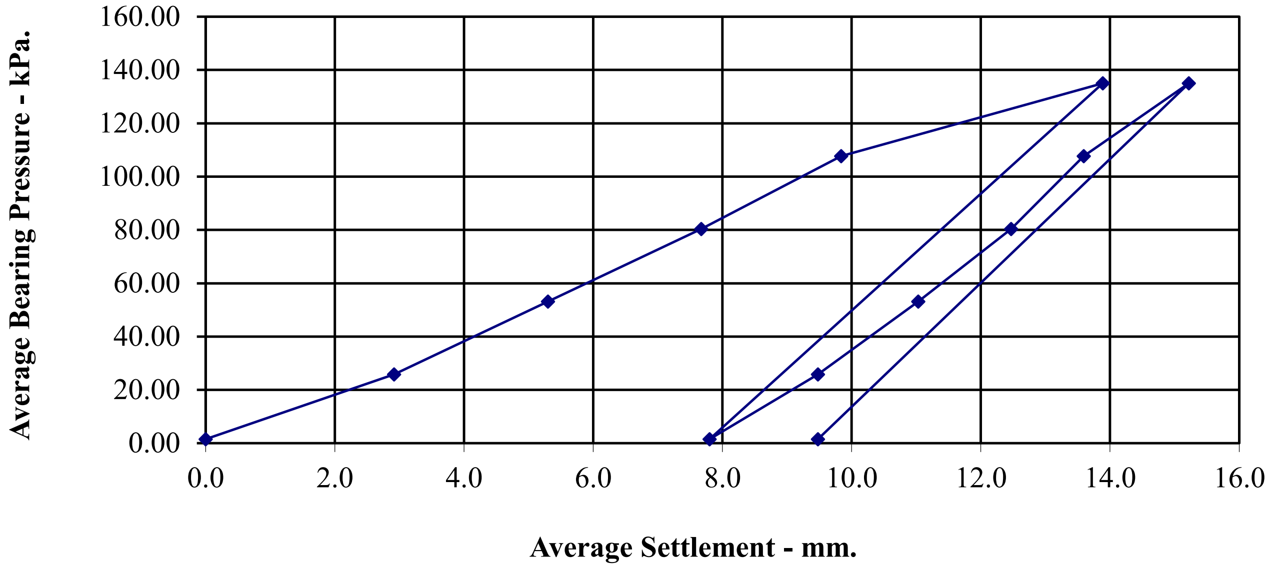
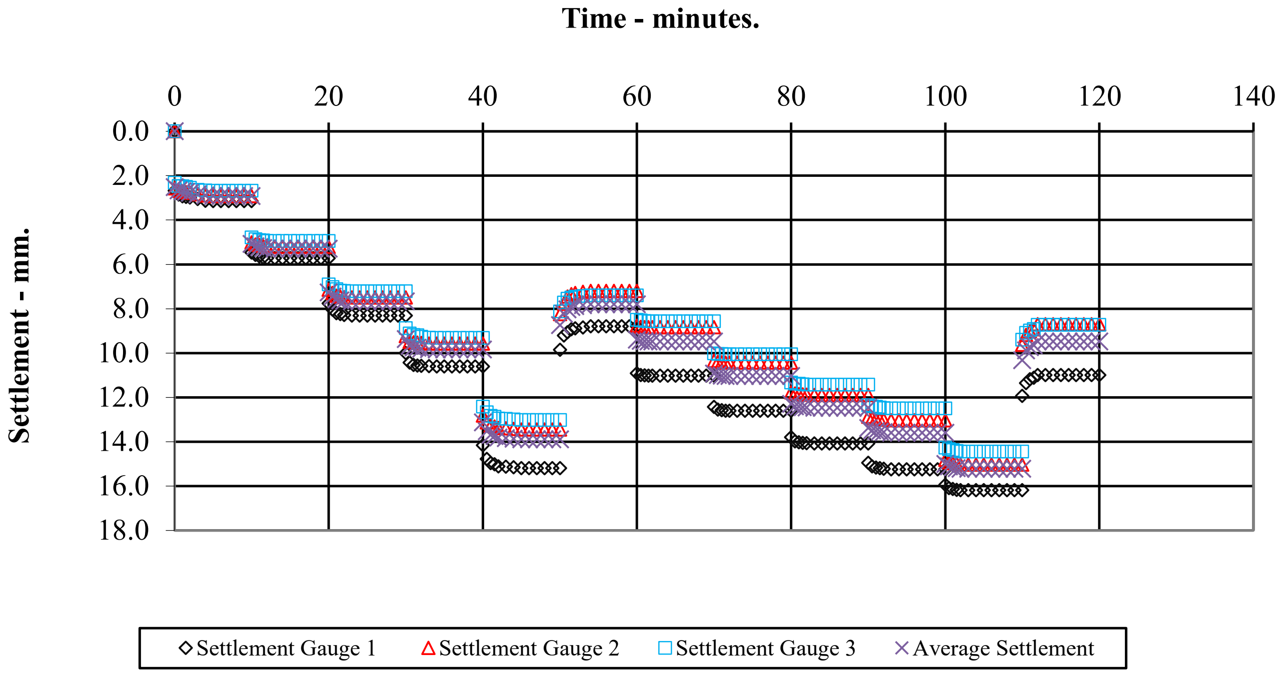
VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test:	**24-Apr-26**	**Test Ref:**	**PBT 6**
Grid Ref:		**Depth (m):**	**0.50**
Layer:		**Conditions:**	**Overcast**

Maximum Applied Pressure (kPa):	**135.02**
Maximum Deformation (mm):	**15.22**
Plate Area (m2):	**0.2922**
Description:	**Brown CLAY**
Ratio of EV2/EV1	**1.87**





UKAS TESTING 4043 / PSL PROFESSIONAL SOILS LABORATORY A PHENNA GROUP COMPANY	**Land off Tinker Lane, Rufforth, York, YO23 3QA**	**Contract No:** **PSL26/3595** **Client Ref:**

Calculation of Equivalent CBR Value from Plate Bearing Test

Design Manual for Roads and Bridges Volume 7: Pavement Design and Maintenance.
HD 25/94: Foundations

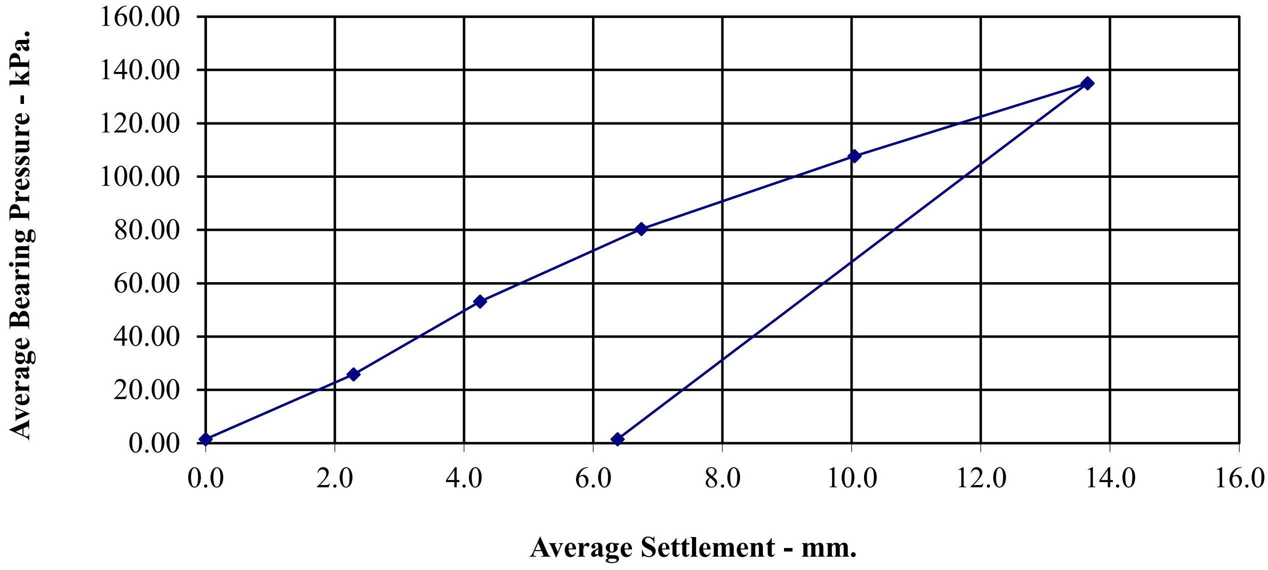
Date of Test	24-Apr-26	
Test Ref	PBT 6	
Depth (m)	0.50	
Grid Ref		
Layer		
Conditions	Overcast	
Description	Brown CLAY	
Maximum Deflection	15.22	mm
Deflection required for CBR value	1.25	mm
Load at 1.25mm	12	kN/m^2
Plate diameter	610	mm
Conversion factor for plate diameter	0.816	
K_{762}(modulus of subgrade reaction) calculated using 1.25mm settlement	7.8	$kN/m^2/mm$
CBR Value	0.3	%

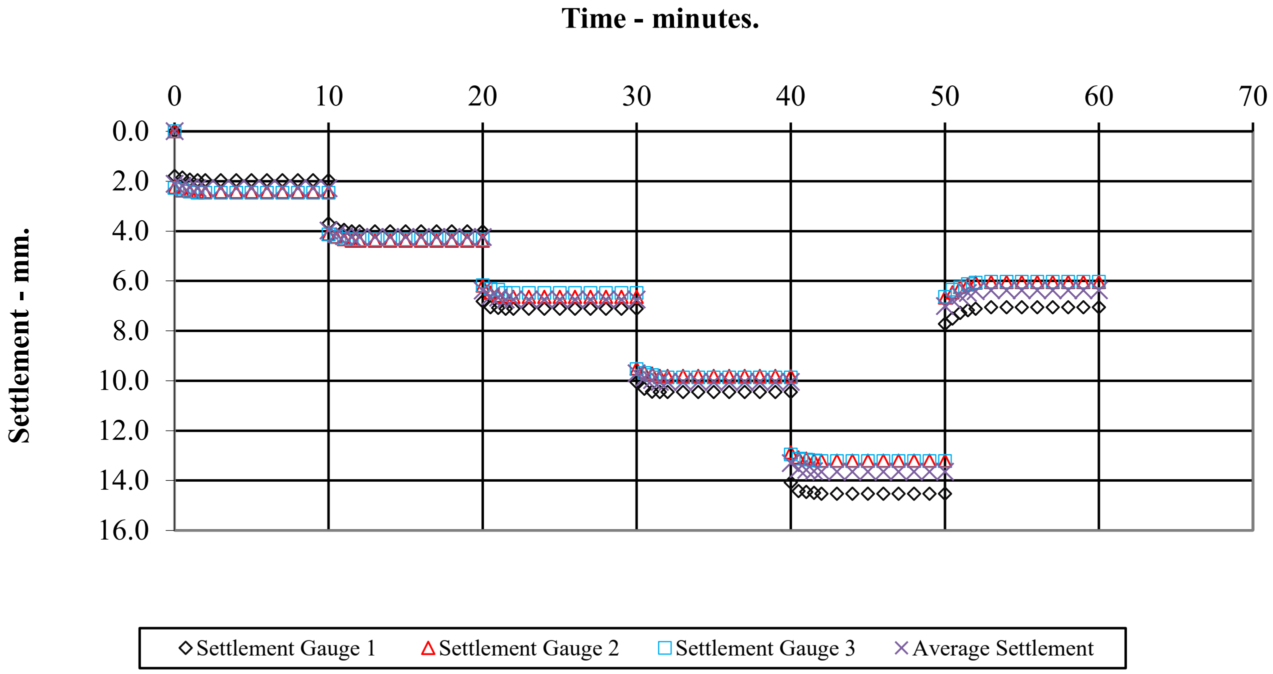
VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test:	**24-Apr-26**	**Test Ref:**	**PBT 7**
Grid Ref:		**Depth (m):**	**0.60**
Layer:		**Conditions:**	**Overcast**

Maximum Applied Pressure (kPa):	**135.02**
Maximum Deformation (mm):	**13.65**
Plate Area (m2):	**0.2922**
Description:	**Brown CLAY**





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HD 25/94: Foundations

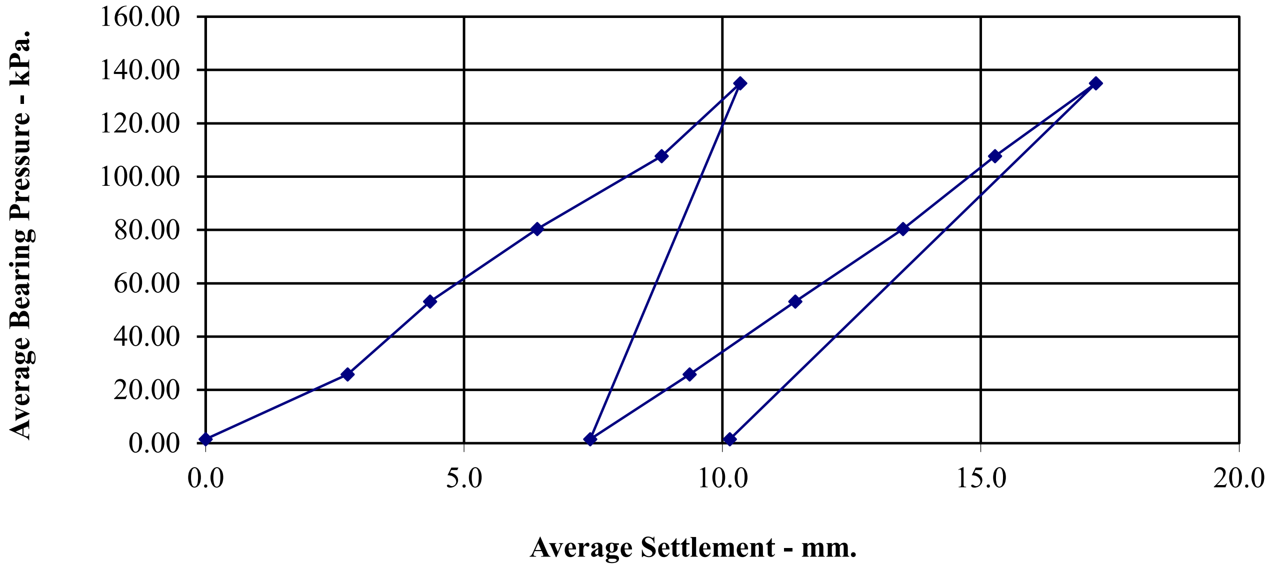
Date of Test	**24-Apr-26**	
Test Ref	**PBT 7**	
Depth (m)	**0.60**	
Grid Ref		
Layer		
Conditions	**Overcast**	
Description	**Brown CLAY**	
Maximum Deflection	**13.65**	**mm**
Deflection required for CBR value	**1.25**	**mm**
Load at 1.25mm	**15**	**kN/m^2**
Plate diameter	**610**	**mm**
Conversion factor for plate diameter	**0.816**	
K_{762}(modulus of subgrade reaction) calculated using 1.25mm settlement	**9.6**	**$kN/m^2/mm$**
CBR Value	**0.5**	**%**

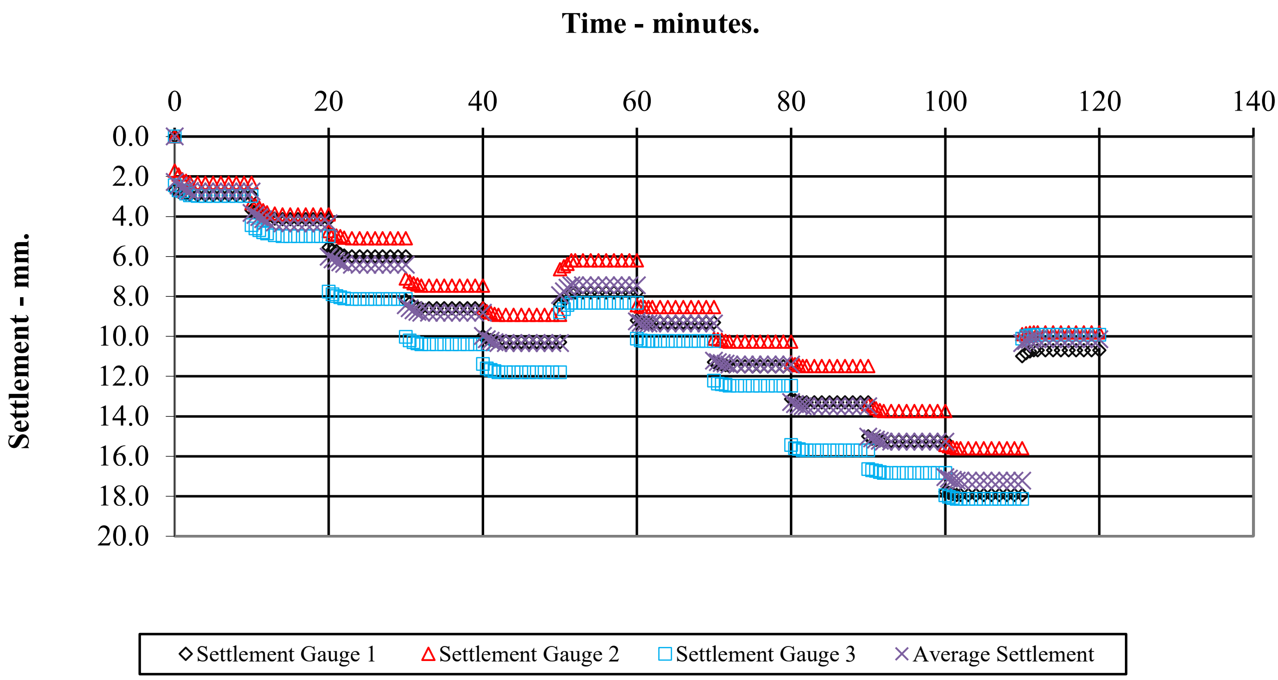
VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test:	**24-Apr-26**	**Test Ref:**	**PBT 8**
Grid Ref:		**Depth (m):**	**0.50**
Layer:		**Conditions:**	**Overcast**

Maximum Applied Pressure (kPa):	**135.02**
Maximum Deformation (mm):	**17.23**
Plate Area (m2):	**0.2922**
Description:	**Brown CLAY**
Ratio of EV2/EV1	**1.06**





UKAS TESTING 4043

PSL PROFESSIONAL SOILS LABORATORY A PHENNA GROUP COMPANY

Land off Tinker Lane, Rufforth, York, YO23 3QA	**Contract No:**
	PSL26/3595
	Client Ref:

Calculation of Equivalent CBR Value from Plate Bearing Test

Design Manual for Roads and Bridges Volume 7: Pavement Design and Maintenance.
HD 25/94: Foundations

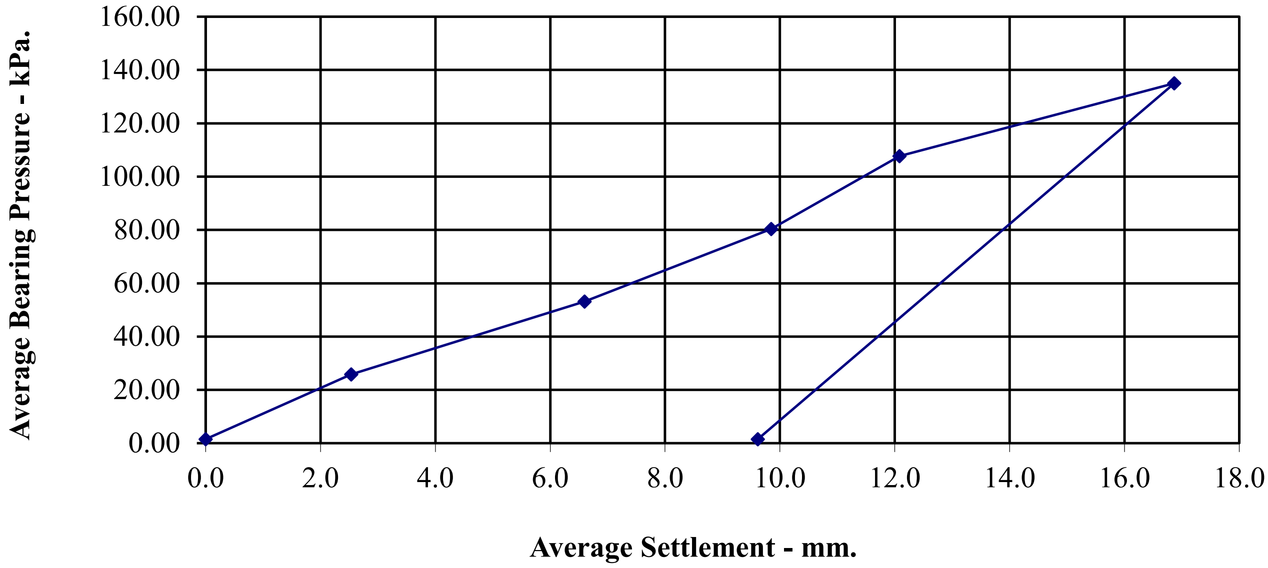
Date of Test	24-Apr-26	
Test Ref	PBT 8	
Depth (m)	0.50	
Grid Ref		
Layer		
Conditions	Overcast	
Description	Brown CLAY	
Maximum Deflection	17.23	mm
Deflection required for CBR value	1.25	mm
Load at 1.25mm	13	kN/m^2
Plate diameter	610	mm
Conversion factor for plate diameter	0.816	
K_{762}(modulus of subgrade reaction) calculated using 1.25mm settlement	8.2	$kN/m^2/mm$
CBR Value	0.4	%

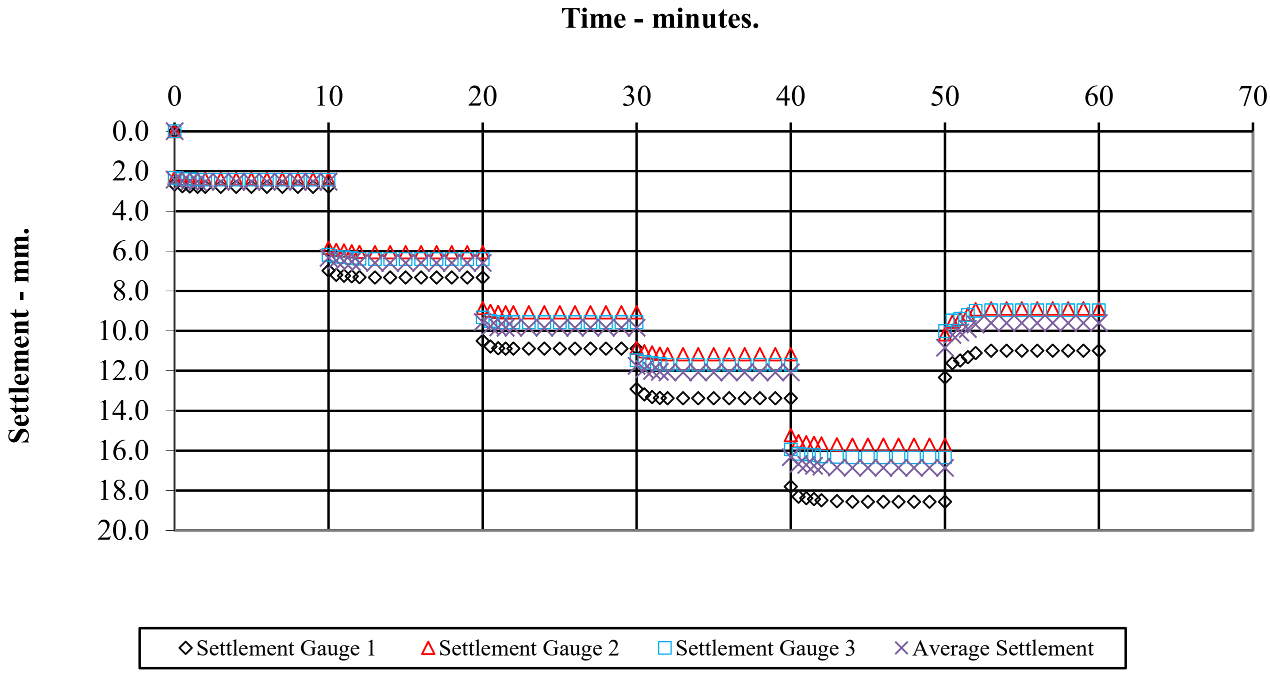
VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test:	**24-Apr-26**	**Test Ref:**	**PBT 9**
Grid Ref:		**Depth (m):**	**0.50**
Layer:		**Conditions:**	**Dry**

Maximum Applied Pressure (kPa):	**135.02**
Maximum Deformation (mm):	**16.87**
Plate Area (m2):	**0.2922**
Description:	**Brown CLAY**





UKAS TESTING 4043 / PSL PROFESSIONAL SOILS LABORATORY A PHENNA GROUP COMPANY	**Land off Tinker Lane, Rufforth, York, YO23 3QA**	**Contract No:** **PSL26/3595** **Client Ref:**

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HD 25/94: Foundations

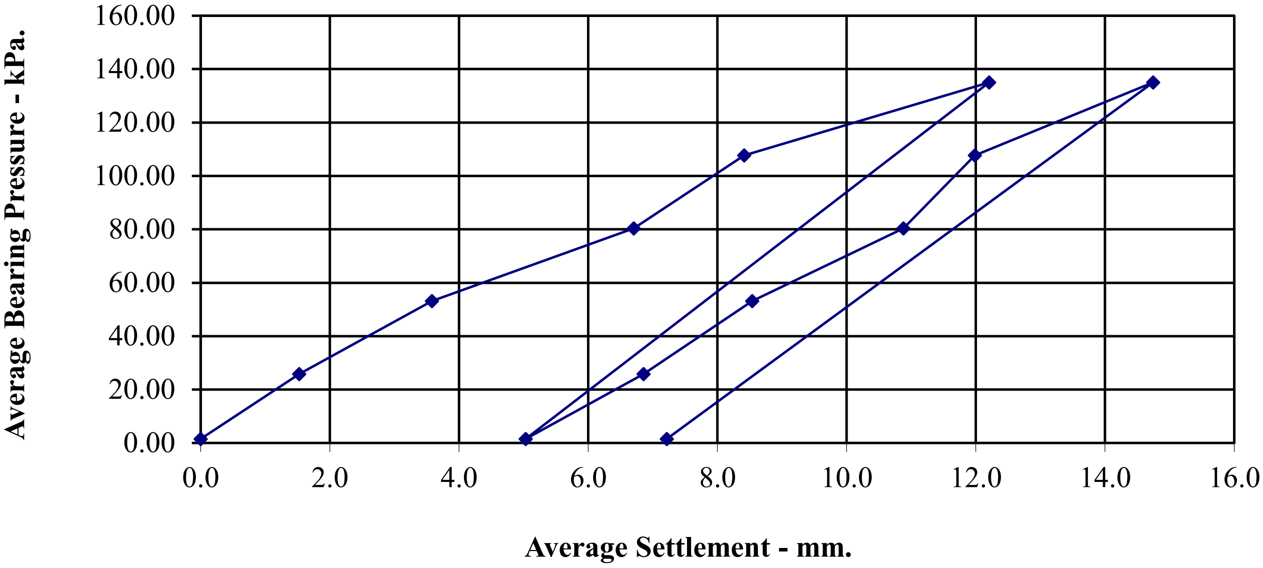
Date of Test	24-Apr-26	
Test Ref	PBT 9	
Depth (m)	0.50	
Grid Ref		
Layer		
Conditions	Dry	
Description	Brown CLAY	
Maximum Deflection	16.87	mm
Deflection required for CBR value	1.25	mm
Load at 1.25mm	13	kN/m^2
Plate diameter	610	mm
Conversion factor for plate diameter	0.816	
K_{762}(modulus of subgrade reaction) calculated using 1.25mm settlement	8.8	$kN/m^2/mm$
CBR Value	0.4	%

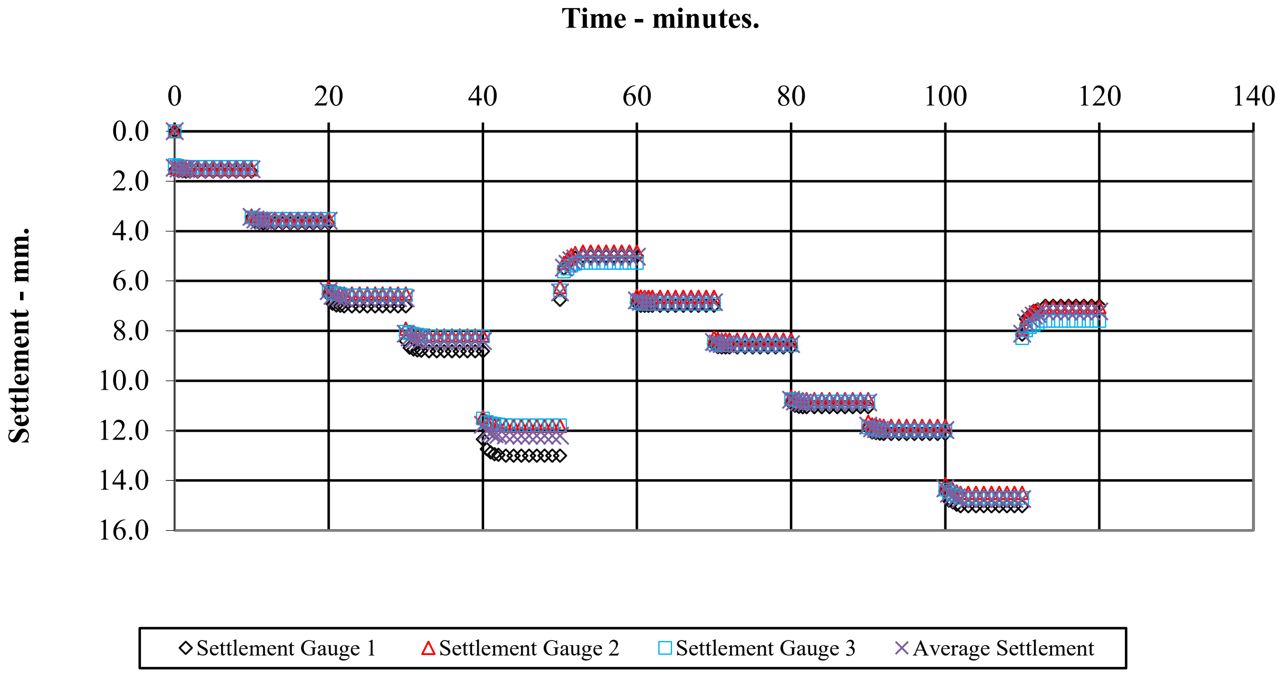
VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test:	**24-Apr-26**	**Test Ref:**	**PBT 10**
Grid Ref:		**Depth (m):**	**0.50**
Layer:		**Conditions:**	**Dry**

Maximum Applied Pressure (kPa):	**135.02**
Maximum Deformation (mm):	**14.74**
Plate Area (m2):	**0.2922**
Description:	**Brown CLAY**
Ratio of EV2/EV1	**1.26**





PSL PROFESSIONAL SOILS LABORATORY A PHENNA GROUP COMPANY	**Land off Tinker Lane, Rufforth, York, YO23 3QA**	**Contract No:** **PSL26/3595** **Client Ref:**

Calculation of Equivalent CBR Value from Plate Bearing Test

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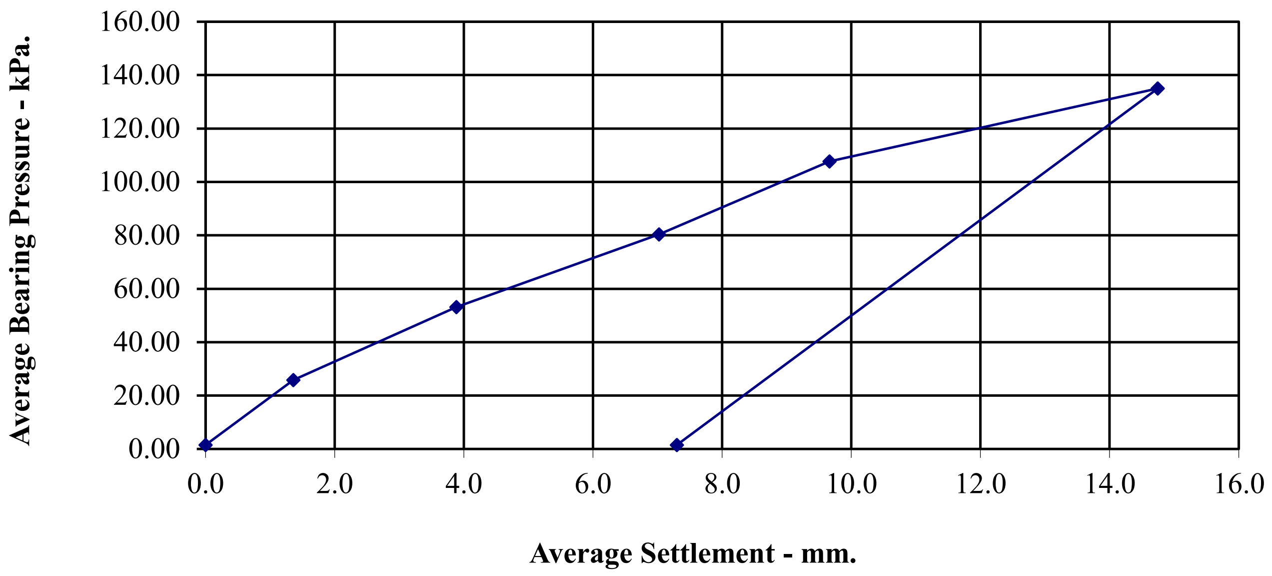
Date of Test	24-Apr-26	
Test Ref	PBT 10	
Depth (m)	0.50	
Grid Ref		
Layer		
Conditions	Dry	
Description	Brown CLAY	
Maximum Deflection	14.74	mm
Deflection required for CBR value	1.25	mm
Load at 1.25mm	21	kN/m^2
Plate diameter	610	mm
Conversion factor for plate diameter	0.816	
K_{762}(modulus of subgrade reaction) calculated using 1.25mm settlement	14.0	$kN/m^2/mm$
CBR Value	0.9	%

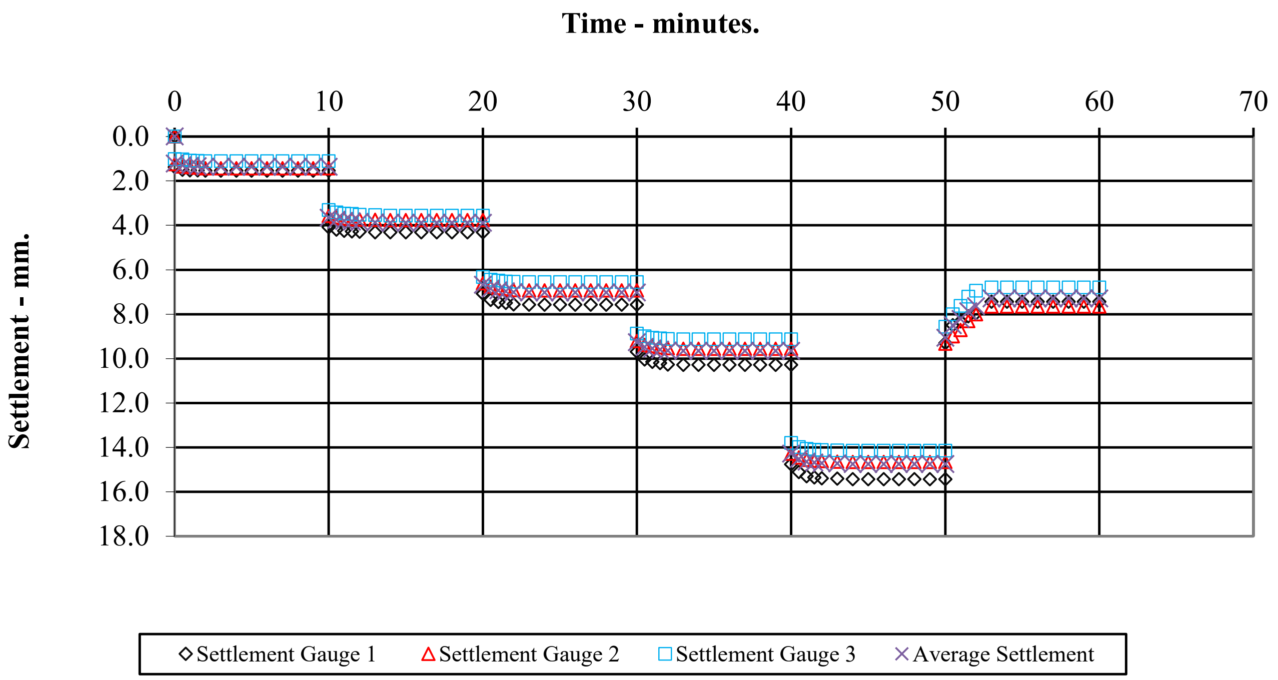
VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test:	**24-Apr-26**	**Test Ref:**	**PBT 11**
Grid Ref:		**Depth (m):**	**0.40**
Layer:		**Conditions:**	**Dry**

Maximum Applied Pressure (kPa):	**135.02**
Maximum Deformation (mm):	**14.75**
Plate Area (m2):	**0.2922**
Description:	**Brown CLAY**





UKAS TESTING 4043 / PSL PROFESSIONAL SOILS LABORATORY A PHENNA GROUP COMPANY	**Land off Tinker Lane, Rufforth, York, YO23 3QA**	**Contract No:** **PSL26/3595** **Client Ref:**

Calculation of Equivalent CBR Value from Plate Bearing Test

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HD 25/94: Foundations

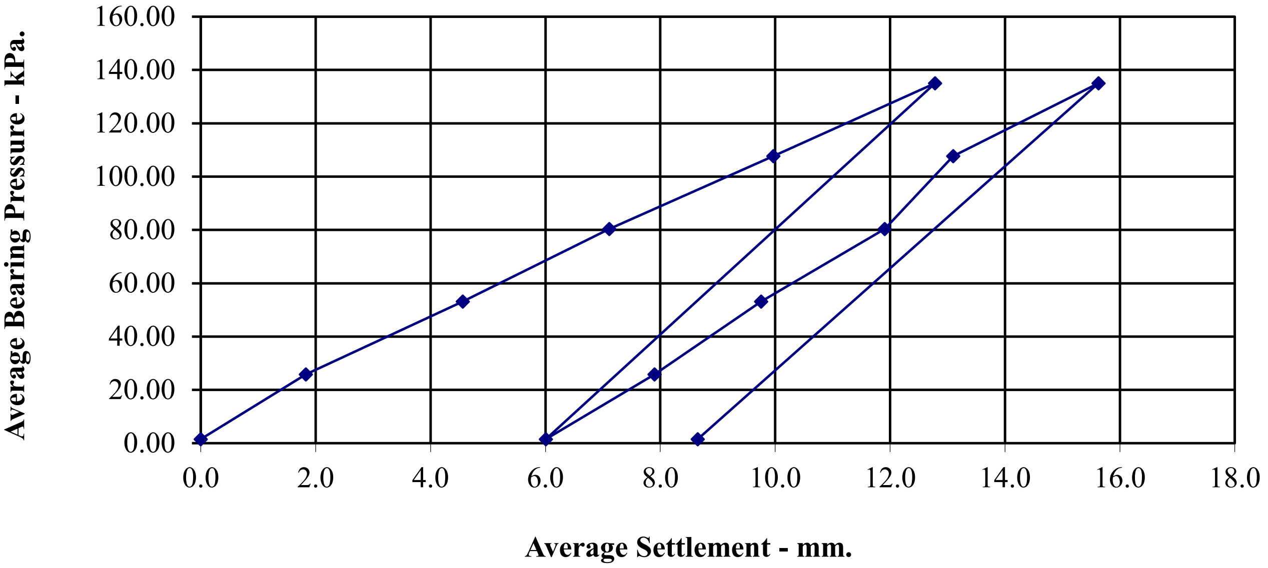
Date of Test	24-Apr-26	
Test Ref	PBT 11	
Depth (m)	0.40	
Grid Ref		
Layer		
Conditions	Dry	
Description	Brown CLAY	
Maximum Deflection	14.75	mm
Deflection required for CBR value	1.25	mm
Load at 1.25mm	24	kN/m^2
Plate diameter	610	mm
Conversion factor for plate diameter	0.816	
K_{762}(modulus of subgrade reaction) calculated using 1.25mm settlement	15.6	$kN/m^2/mm$
CBR Value	1.1	%

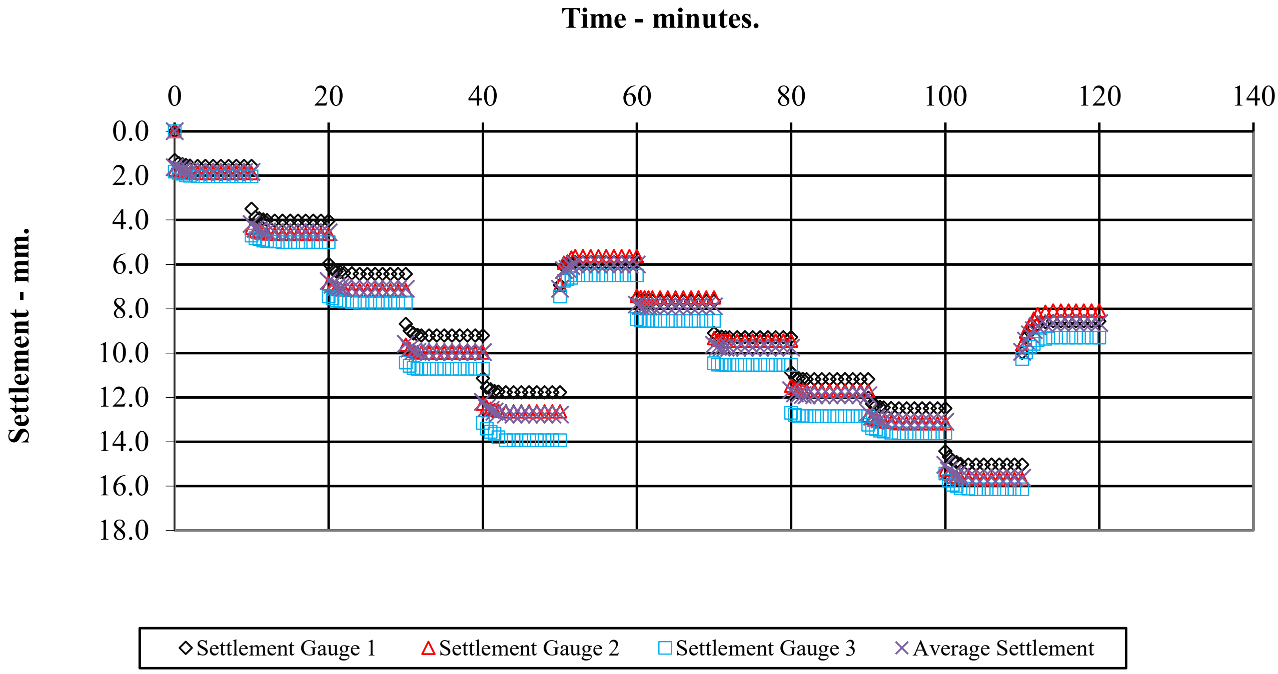
VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test:	**24-Apr-26**	**Test Ref:**	**PBT 12**
Grid Ref:		**Depth (m):**	**0.40**
Layer:		**Conditions:**	**Dry**

Maximum Applied Pressure (kPa):	**135.02**
Maximum Deformation (mm):	**15.63**
Plate Area (m2):	**0.2922**
Description:	**Brown CLAY**
Ratio of EV2/EV1	**1.33**





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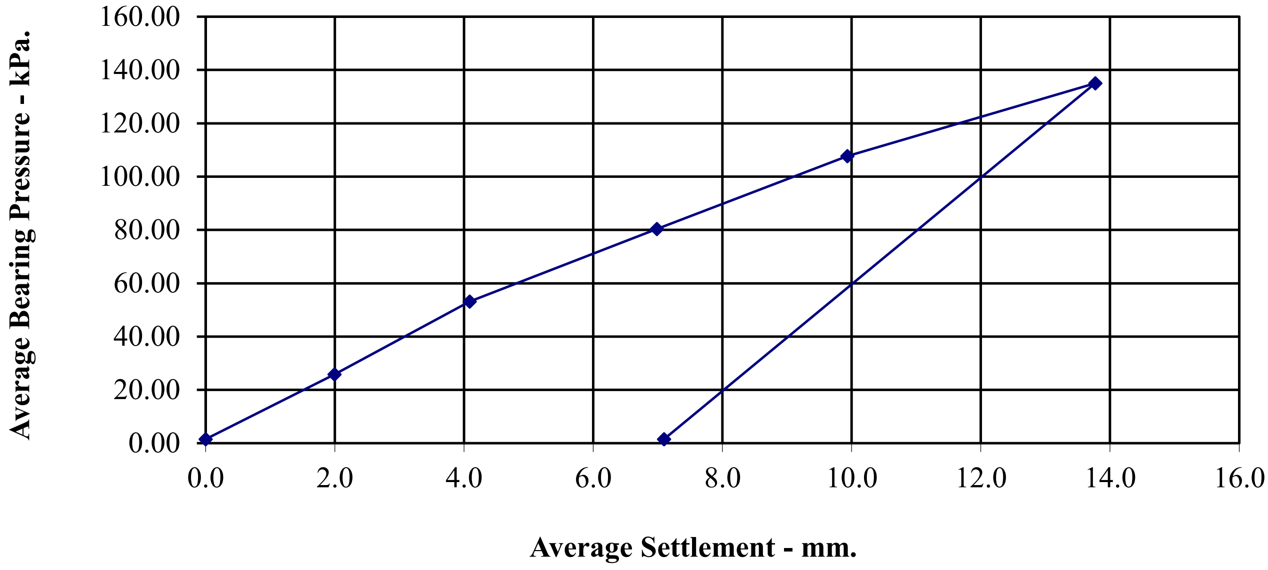
Date of Test	**24-Apr-26**	
Test Ref	**PBT 12**	
Depth (m)	**0.40**	
Grid Ref		
Layer		
Conditions	**Dry**	
Description	**Brown CLAY**	
Maximum Deflection	**15.63**	**mm**
Deflection required for CBR value	**1.25**	**mm**
Load at 1.25mm	**18**	**kN/m^2**
Plate diameter	**610**	**mm**
Conversion factor for plate diameter	**0.816**	
K_{762}(modulus of subgrade reaction) calculated using 1.25mm settlement	**11.8**	**$kN/m^2/mm$**
CBR Value	**0.7**	**%**

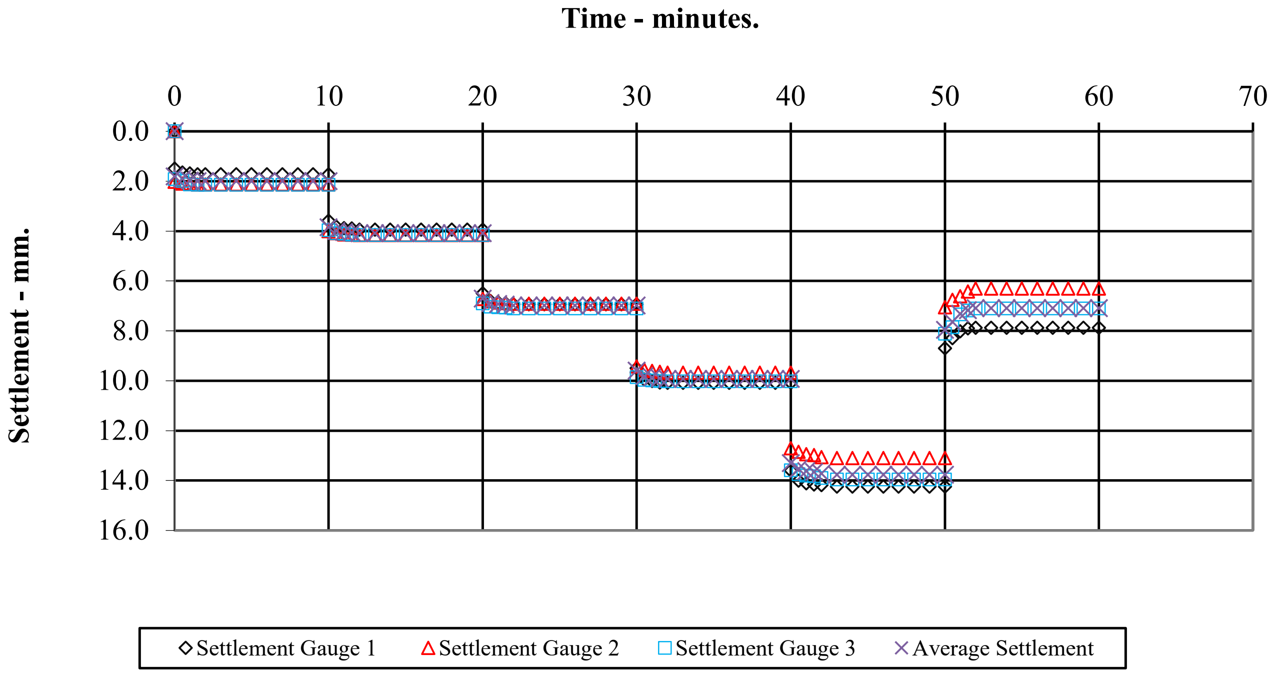
VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test:	**24-Apr-26**	**Test Ref:**	**PBT 13**
Grid Ref:		**Depth (m):**	**0.40**
Layer:		**Conditions:**	**Dry**

Maximum Applied Pressure (kPa):	**135.02**
Maximum Deformation (mm):	**13.77**
Plate Area (m2):	**0.2922**
Description:	**Brown CLAY**





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Date of Test	24-Apr-26	
Test Ref	PBT 13	
Depth (m)	0.40	
Grid Ref		
Layer		
Conditions	Dry	
Description	Brown CLAY	
Maximum Deflection	13.77	mm
Deflection required for CBR value	1.25	mm
Load at 1.25mm	17	kN/m^2
Plate diameter	610	mm
Conversion factor for plate diameter	0.816	
K_{762}(modulus of subgrade reaction) calculated using 1.25mm settlement	10.9	$kN/m^2/mm$
CBR Value	0.6	%